THE CONORMAL SEQUENCE OF AN IMMERSION OF ALGEBRAIC SPACES WITH A LEFT INVERSE SPLITS ETALE LOCALLY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1179-1191 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1193-1201, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The relative conormal-differentials sequence for immersions of algebraic spaces (same chapter, lemma-differentials-relative-immersion, collected as S3-spaces-differentials-rel-immersion) and the etale localization of differentials and conormal sheaves (same chapter); cited inputs with the splitting claim written in the retained span.

Source reference. Original lemma 05ZA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1145–1155. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CO: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 432–441. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $i : Z \to X$ be an immersion of algebraic spaces over B, and assume i (étale locally) has a left inverse. Then the canonical sequence*

$$0 \to \mathcal{C}_{Z/X} \to i^*\Omega_{X/B} \to \Omega_{Z/B} \to 0$$

of Lemma 05ZA is (étale locally) split exact.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Clarification: we claim that if $g : X \to Z$ is a left inverse of i over B, then i^*c_g is a right inverse of the map $i^*\Omega_{X/B} \to \Omega_{Z/B}$. Having said this, the result follows from the corresponding result for morphisms of schemes by étale localization, see Lemmas 04CU and 04CO. □